

Corporate Bankruptcy Risk Prediction with Machine Learning

Comparing Interpretable Logistic Regression
and Tree-Based Models

Berke Pehlivan

Machine Learning for Finance

University of Bonn

Academic Year 2025/2026

Abstract

This paper compares classification models for identifying failed company-year observations from financial statement variables. The data contain 78,682 observations from 8,971 American companies between 1999 and 2018. Because firms appear in several years, companies rather than individual rows are assigned to the training and test samples. The comparison includes a majority-class baseline, Logistic Regression with regularization, a Decision Tree, Random Forest, Gradient Boosting, and a PCA plus Logistic Regression extension. Evaluation focuses on the minority failed class. The majority baseline reaches 92.5% test accuracy but detects no failures, showing why accuracy is misleading. Logistic Regression has the highest failed recall among the key models but produces many false alarms. Random Forest has the highest test average precision and failed-class F1-score, while Gradient Boosting has the highest balanced accuracy. Threshold analysis shows that greater recall comes with lower precision and a larger review burden. PCA retains much of the predictor variance but does not clearly improve validation performance and reduces interpretability. The results support machine learning as a structured screening exercise, not as a causal or production-ready bankruptcy system.

Contents

1	Introduction	1
1.1	Financial motivation	1
1.2	Research question and contribution	1
2	Data	2
2.1	Dataset and unit of observation	2
2.2	Target and financial predictors	3
2.3	Class imbalance and descriptive evidence	3
3	Methodology	5
3.1	Company-level training, validation, and test design	5
3.2	Majority-class baseline	5
3.3	Logistic Regression and regularization	5
3.4	Tree-based models	6
3.5	PCA extension	6
4	Evaluation Strategy	7
4.1	Why accuracy is insufficient	7
4.2	Ranking and failed-class metrics	7
4.3	Threshold analysis	8
5	Results	8
5.1	Final-test model comparison	8

5.2	Classification trade-offs	9
5.3	Precision–recall analysis	10
6	Model Interpretation	11
6.1	Logistic Regression coefficients	11
6.2	Tree-based feature importance	12
7	PCA Results	12
8	Critical Reflection and Limitations	13
8.1	Data and target limitations	13
8.2	Validation and modelling limitations	14
8.3	Decision and practical limitations	14
8.4	Priorities for future work	15
9	Conclusion	15
A	Additional Model Results	17
B	Threshold Analysis	18
C	PCA Diagnostics	19
D	Reproducibility and Code Structure	20

1 Introduction

1.1 Financial motivation

Corporate bankruptcy is an important problem for lenders, investors, auditors, and risk managers because a firm’s failure can impose losses on creditors and shareholders and can disrupt relationships with employees, suppliers, and other firms. Financial statements provide a structured record of a company’s assets, liabilities, revenues, expenses, and profitability. These variables may contain signals of financial distress before failure occurs, which makes bankruptcy prediction a natural application of statistical learning in finance. Early research established that combinations of accounting variables can help distinguish distressed from healthy firms (Altman, 1968), while later work showed how bankruptcy risk can be studied in a probabilistic classification framework (Ohlson, 1980).

Despite this long research history, bankruptcy prediction remains difficult. Failure is a rare outcome, so a model can achieve high overall accuracy by predicting that almost every observation remains alive. Such a model may appear successful while failing to identify the firms that matter most for a distress screening exercise. In addition, financial variables are often strongly correlated, measured on very different scales, and affected by firm size. Relationships between accounting variables and failure may also be nonlinear or depend on interactions that are not represented by a simple linear model. Finally, false negatives and false positives do not generally have the same economic cost: missing a failed firm may be costly, but flagging too many healthy firms can make a screening system impractical.

These features make model evaluation as important as model fitting. Accuracy is reported for completeness, but it cannot be the main criterion in an imbalanced setting. This study therefore emphasizes precision and recall for the failed class, the failed-class F1-score, balanced accuracy, the area under the receiver operating characteristic curve (ROC-AUC), and average precision, reported in the project as PR-AUC. Precision–recall analysis is especially informative when the negative class greatly outnumbers the positive class, because it makes the false-positive burden visible rather than allowing the many true negatives to dominate the assessment (Davis and Goadrich, 2006). The probability threshold is also treated as a decision choice rather than assuming that the default value of 0.50 is economically optimal.

1.2 Research question and contribution

This paper investigates the following research question:

How effectively can interpretable and tree-based machine-learning models distinguish failed from alive company-year observations for previously unseen companies, and what trade-offs arise between failure detection and false alarms?

The analysis uses the American corporate bankruptcy dataset introduced by Lombardo et al. (2022). The working sample contains 78,682 company-year observations from 8,971 anonymized companies between 1999 and 2018, with 18 financial predictors and a binary failed/alive target. Because companies may appear in multiple years, observations are split at the company level. This prevents the same company from appearing in both training and final-test data and reduces the risk that persistent company characteristics create an artificially favorable evaluation. The

design tests generalization to unseen companies; it is not presented as a chronological forecasting backtest.

The empirical comparison begins with a majority-class baseline, which demonstrates why accuracy alone is misleading. Logistic Regression provides the main interpretable benchmark, and L1- and L2-regularized variants examine whether coefficient shrinkage improves generalization when predictors are correlated. A Decision Tree, Random Forest, and Gradient Boosting model then test whether nonlinear rules and interactions improve predictive performance. These models represent a progression from a transparent parametric classifier to more flexible ensemble methods, reflecting the interpretability–flexibility and bias–variance trade-offs emphasized in statistical learning (James et al., 2023). A PCA plus Logistic Regression specification is included as a dimension-reduction extension, not as the main model.

The paper makes three practical contributions within the scope of a course project. First, it constructs a reproducible and leakage-aware comparison of interpretable and tree-based classifiers. Second, it evaluates the models using metrics that focus directly on the minority failed class and uses an untouched final test set for the principal comparison. Third, it connects statistical performance to the financial screening problem by examining confusion-matrix counts, validation-based threshold trade-offs, Logistic Regression coefficients, and tree-based feature importance. The resulting estimates are interpreted as predictive associations rather than causal effects. The study is therefore an empirical model comparison and not a claim that the fitted models are ready for automated credit or investment decisions.

2 Data

2.1 Dataset and unit of observation

The project uses the American corporate bankruptcy dataset published by Lombardo et al. (2022). It contains accounting information for public companies listed in the United States. The version used in this project has 78,682 rows covering the years 1999 to 2018. It contains 8,971 anonymized company identifiers, 18 financial predictors, a year variable, and a company status. The data summary reports no missing values and no duplicated rows.

One row represents one company in one year. The same company can therefore appear in several years. This panel-like structure is important for the way the sample is split. A random split of individual rows could place one company’s earlier observations in the training sample and its later observations in the test sample. The model might then benefit from company-specific information that is already present in training. To reduce this risk, the project assigns whole companies to the training or final-test sample. The split design is explained in Section 3.

The source data cover a long period in which both the set of listed companies and the wider economic environment changed. However, the project does not use a time-based holdout. The final test set contains unseen companies, but its years overlap with the years in the training set. The results should therefore be read as evidence about generalization across companies under this sample design, not as a real-time forecast of a later economic period.

2.2 Target and financial predictors

The original status is recorded as **alive** or **failed**. For modeling, failed observations are coded as 1 and alive observations as 0. The failed class is treated as the positive class throughout the analysis. The company identifier and year are kept for splitting and reporting, but they are not used as predictors. This leaves 18 financial inputs, labelled **X1** to **X18** in the source file.

The predictors cover several parts of a company’s financial position and performance. They include current assets, inventory, receivables, market value, and total assets; profitability measures such as EBITDA, EBIT, net income, gross profit, and retained earnings; long-term debt, current liabilities, and total liabilities; and activity measures such as sales, revenue, cost of goods sold, and operating expenses. Depreciation and amortization is also included. The model therefore uses reported financial levels rather than a set of engineered financial ratios. Readable names are retained in the tables and figures so that the model outputs can be discussed in financial terms.

These variables have very different scales and several describe related parts of the same financial statements. For example, revenue and net sales are closely connected, as are assets and liabilities. This matters for Logistic Regression, because coefficient size depends on measurement scale and correlated predictors can make individual coefficients less stable. The Logistic Regression and PCA pipelines therefore standardize the predictors using parameters learned from the training data. Tree-based models do not require this scaling.

2.3 Class imbalance and descriptive evidence

The target is strongly imbalanced. Of the 78,682 company-year observations, 73,462 are alive and 5,220 are failed. The failed class therefore represents only 6.63% of the full sample. Figure 1 shows this difference. A classifier that predicts every observation as alive would be correct in more than 93% of the full sample, although it would detect no failures. This is the main reason why the later evaluation does not rely on accuracy alone.

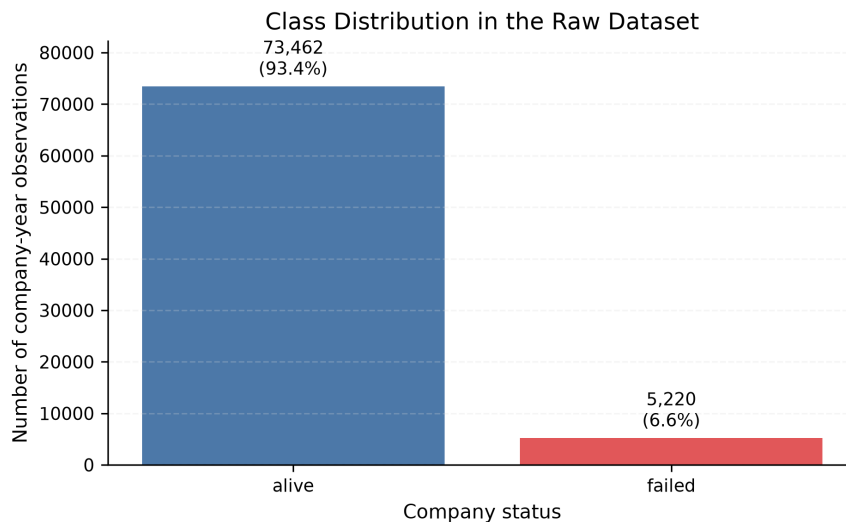


Figure 1: Class distribution in the full dataset. The failed class accounts for 6.6% of company-year observations.

The share of failed observations also changes over time. As shown in Figure 2, the observed

rate rises from 7.16% in 1999 to a peak of 9.40% in 2003 and then generally declines, reaching 1.32% in 2018. This pattern should be interpreted with care. It may reflect changes in economic conditions, but it may also reflect changes in the companies included in the dataset or in the way the status is represented. The figure is used as a description of the sample and not as evidence of a causal time trend.

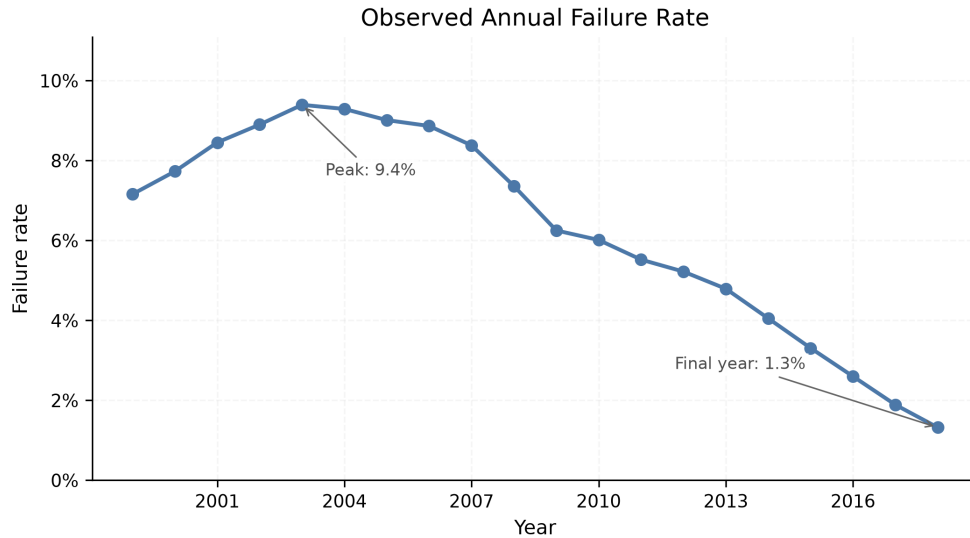


Figure 2: Observed annual failure rate in the full dataset, 1999–2018.

Table 1 compares the medians of six selected financial variables. Medians are used because financial statement values are often skewed and can contain very large observations. Failed observations have lower median market value, net income, current assets, and retained earnings. They have higher median long-term debt and total liabilities. The signs are financially plausible as descriptive patterns, but they do not show that these variables cause failure. They are simple comparisons made before controlling for the other predictors.

Table 1: Selected financial-variable medians by company status

Financial variable	Alive median	Failed median
Market value	240.90	117.80
Net income	2.07	-3.33
Total long-term debt	7.09	18.44
Current assets	102.92	75.87
Total liabilities	80.74	97.04
Retained earnings	-0.17	-25.51

Notes: Values are shown in the units reported by the source dataset. The table is descriptive and does not imply causal effects.

3 Methodology

3.1 Company-level training, validation, and test design

The full sample is first divided into training and final-test data with a grouped random split. The company identifier is used as the group, so every row from a company is assigned to the same side of the split. The split places 80% of companies in training and 20% in the final test set. A fixed random seed of 42 is used so that the result can be reproduced. The training set contains 62,988 rows from 7,176 companies, while the final test set contains 15,694 rows from 1,795 companies. The failure rates are 6.42% and 7.50%, respectively.

A second company-level split is created inside the training set. It reserves 20% of the training companies for validation. This produces a model-fitting sample of 50,323 rows from 5,740 companies and a validation sample of 12,665 rows from 1,436 companies. Hyperparameters are compared on this validation sample using PR-AUC. The same validation predictions are later used for the threshold analysis. The final test set is not used for either choice.

The saved models remain fitted on the inner model-fitting sample and are then evaluated on the final test set. In other words, the validation rows are not added back for a final refit. This is a conservative design because fewer rows are used to fit the final models. A common alternative would be to refit the selected specification on all training rows before the final evaluation. That alternative was not used here, and this point is considered again in the limitations section.

All random model components use the same seed. The implementation uses scikit-learn pipelines and estimators (Pedregosa et al., 2011). Pipelines are important for Logistic Regression and PCA because standardization is learned only from the fitting data and then applied to validation or test observations. This avoids using information from held-out observations during preprocessing.

3.2 Majority-class baseline

The first model is a majority-class baseline. It always predicts the most common training label, which is `alive`. The baseline does not try to learn a relationship between the financial variables and the target. Its purpose is to give a simple reference point. Because alive observations are very common, this rule achieves high accuracy while its recall for failures is zero. Any useful bankruptcy classifier should therefore be judged by more than whether it beats the baseline’s accuracy.

3.3 Logistic Regression and regularization

Logistic Regression is the main interpretable model. For observation i , the model links the probability of failure p_i to the financial predictors through

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{18} x_{i18}. \quad (1)$$

The left-hand side is the log-odds of failure. A positive coefficient is linked to higher predicted failure log-odds when the other variables are held fixed; a negative coefficient is linked to lower predicted failure log-odds. The coefficients are not direct percentage-point changes in probability, and they are not interpreted as causal effects.

Each Logistic Regression specification is placed after a standardization step. The predictors therefore have mean zero and standard deviation one in the model-fitting sample. This makes the regularization penalty comparable across variables with different units. Balanced class weights give more importance to errors on the minority failed class during fitting.

Three Logistic Regression specifications are compared. The main benchmark uses an L2 penalty with fixed $C = 1$. The parameter C is the inverse of the regularization strength, so a smaller value means stronger shrinkage. Separate L1 and L2 models compare $C \in \{0.01, 0.1, 1, 10\}$ using validation PR-AUC. Both regularized models select $C = 0.1$. L1 regularization can set some coefficients exactly to zero, while L2 regularization usually keeps all variables but shrinks their coefficients. These methods can reduce variance when predictors are correlated, although they add some bias (James et al., 2023).

3.4 Tree-based models

Tree-based models are included because they can represent nonlinear patterns and interactions without manually specifying them. A Decision Tree repeatedly divides the predictor space into smaller regions. Its depth and minimum leaf size control complexity. A deep tree may fit the training data closely but can have high variance. The selected tree has a maximum depth of 5 and at least 50 observations in each leaf. Balanced class weights are used during fitting.

Random Forest reduces the variance of a single tree by averaging many trees fitted on resampled data. It also considers a random subset of predictors at each split, which makes the trees less similar to each other (Breiman, 2001). The selected model uses 300 trees, no fixed maximum depth, at least 50 observations per leaf, and the square root of the available predictors at each split. Class weights are balanced separately within each bootstrap sample.

Gradient Boosting also combines many trees, but it builds them in sequence. Each new tree is added to improve the errors left by the existing ensemble (Friedman, 2001). The learning rate controls how much each new tree changes the model. The selected specification uses 150 trees, a learning rate of 0.05, a maximum tree depth of 3, and at least 100 observations per leaf. The scikit-learn Gradient Boosting estimator does not have a direct class-weight argument, so balanced observation weights are supplied during fitting.

The parameter grids are intentionally small. They cover the main complexity controls without turning the project into a large search exercise. A compact specification table is provided in Appendix A.

3.5 PCA extension

Principal Component Analysis (PCA) is included as a separate extension. PCA replaces the original standardized predictors with linear combinations called principal components. The first component captures the largest possible share of variation in the predictors, and each later component captures another orthogonal direction of variation (Jolliffe and Cadima, 2016). This can summarize correlated financial variables with fewer inputs.

The PCA pipeline first standardizes the 18 predictors, then applies PCA, and finally fits a balanced Logistic Regression with $C = 1$. The tested component counts are 2, 3, 5, 8, 10, 12, 15, and 18. They are compared on the same type of internal company-level validation split used

for the main models. The PCA results are kept as validation results and are not presented as final-test results.

PCA is useful for compression, but it has two important costs in this setting. First, it is unsupervised: components are chosen to explain variation in the predictors, not to separate failed and alive observations. A component with high explained variance is therefore not automatically useful for classification. Second, each component mixes several financial variables. This makes the model harder to explain than Logistic Regression fitted directly on current assets, income, debt, and the other original variables.

4 Evaluation Strategy

4.1 Why accuracy is insufficient

Accuracy is the share of all observations classified correctly. Although it is easy to understand, it can be misleading when one class is much more common than the other. In this dataset, a model can obtain high accuracy by predicting almost every observation as alive. Such a model would still fail at the main task of identifying firms at risk of failure. The majority-class baseline is therefore included to make this problem visible.

The confusion matrix provides a clearer view of the errors. Failed firms are treated as the positive class. A true positive is a failed observation that the model detects, while a false negative is a failed observation that it misses. A false positive is an alive observation flagged as failed, and a true negative is an alive observation correctly classified as alive. In a screening setting, false negatives represent missed failures and false positives represent false alarms. Both matter, but their costs are not assumed to be equal.

Balanced accuracy is also reported. It is the average of the recall for the failed class and the recall for the alive class:

$$\text{Balanced accuracy} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right). \quad (2)$$

Unlike ordinary accuracy, this measure gives equal weight to both classes.

4.2 Ranking and failed-class metrics

The evaluation gives special attention to the minority failed class. Failed precision measures how many observations predicted as failed actually failed:

$$\text{Precision}_{\text{failed}} = \frac{TP}{TP + FP}. \quad (3)$$

Failed recall measures how many actual failures were detected:

$$\text{Recall}_{\text{failed}} = \frac{TP}{TP + FN}. \quad (4)$$

Their harmonic mean is the failed-class F1-score:

$$F1_{\text{failed}} = 2 \frac{\text{Precision}_{\text{failed}} \times \text{Recall}_{\text{failed}}}{\text{Precision}_{\text{failed}} + \text{Recall}_{\text{failed}}}. \quad (5)$$

Recall is important when missed failures are a concern, while precision shows how many false alarms are produced. The F1-score summarizes their balance, but it does not attach a financial cost to either type of error.

Two threshold-independent ranking measures are reported. ROC-AUC measures how well a model ranks a randomly chosen failed observation above a randomly chosen alive observation across all thresholds. However, an ROC curve can appear quite favourable when the negative class is much larger. Precision–recall analysis focuses more directly on performance for the positive minority class and is therefore especially useful here (Davis and Goadrich, 2006).

In the project output, the measure labelled PR-AUC is calculated using the average precision implementation in scikit-learn (Pedregosa et al., 2011). Average precision summarizes the precision–recall relationship by weighting precision at each level by the increase in recall. This naming detail is stated explicitly because average precision is not exactly the same as applying the trapezoidal rule to a precision–recall curve.

The ranking measures use predicted scores across many possible cut-offs. In contrast, accuracy, precision, recall, F1, and the confusion matrix require one classification threshold and a set of predicted labels. The two groups of measures answer different questions: ranking measures assess ordering quality, while threshold-based measures describe the decisions made at a chosen cut-off.

4.3 Threshold analysis

The default classification threshold of 0.50 is not necessarily the most appropriate choice for bankruptcy screening. Lowering the threshold usually detects more failed firms, but it also creates more false alarms. Raising it usually improves precision at the cost of lower recall. Threshold choice is therefore a practical trade-off rather than a purely technical decision.

Thresholds from 0.01 to 0.99 are examined using only the internal validation predictions for Logistic Regression, Random Forest, and Gradient Boosting. The analysis considers three rules: the threshold with the highest failed-class F1-score, the highest-precision threshold that still reaches at least 60% failed recall, and the corresponding rule for at least 80% failed recall. This shows how the preferred cut-off changes when detecting failures receives more weight.

The final test set is not used to choose a threshold. To keep the main model comparison consistent, the reported final-test classifications use each estimator’s default decision rule, equivalent to a 0.50 probability cut-off for the models considered. The validation threshold exercise is presented as a sensitivity analysis, not as a deployed policy. A real decision rule would also require calibrated probabilities, explicit financial costs, and evidence from a later or external sample. These elements are outside the scope of this project.

5 Results

5.1 Final-test model comparison

Figure 3 compares the majority-class baseline with Logistic Regression, Random Forest, and Gradient Boosting on the held-out test set. These models are shown because they make the main performance trade-offs easy to see. The complete saved results also contain the regularized logistic models and the single Decision Tree. Gradient Boosting is the model selected by validation

PR-AUC, while the final test set is used only for evaluation.

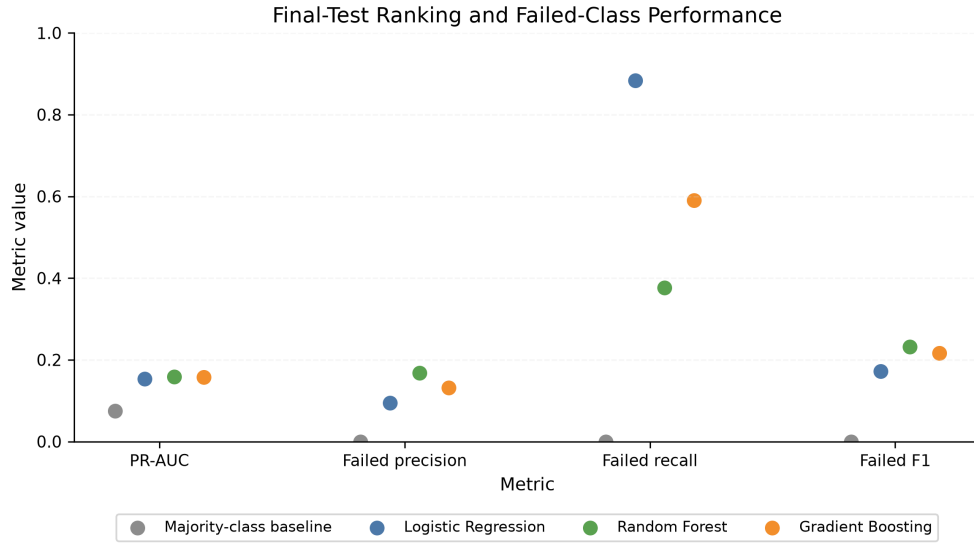


Figure 3: Final-test performance for the failed class. PR-AUC refers to average precision. The values come from the fixed held-out test set.

The majority baseline reaches 0.925 accuracy, but it detects none of the 1,177 failed observations. Its failed-class precision, recall, and F1-score are all zero. This confirms that accuracy alone gives the wrong impression in this imbalanced problem.

No model is best on every measure. Logistic Regression has the highest failed recall at 0.884, but its failed precision is only 0.095. Random Forest has the numerically highest observed final-test PR-AUC among the models shown at 0.159 and the highest failed-class F1-score at 0.232. This is a test-set observation, not a retroactive model-selection rule. Gradient Boosting has the highest balanced accuracy at 0.638 and an intermediate recall of 0.590. The ROC-AUC values of the three models are fairly close, ranging from 0.688 to 0.699. The results therefore support a comparison of trade-offs rather than one simple claim that a single model wins.

5.2 Classification trade-offs

Table 2 translates the final-test predictions into screening outcomes. Logistic Regression detects 1,041 failures and misses only 136. However, it also creates 9,906 false alarms. This explains why its recall is high while its precision and overall accuracy are low.

Table 2: Final-test classification outcomes at the default decision rule

Model	Correct alive	False alarms	Missed failures	Detected failures
Majority baseline	14,517	0	1,177	0
Logistic Regression	4,611	9,906	136	1,041
Random Forest	12,327	2,190	735	442
Gradient Boosting	9,954	4,563	482	695

Random Forest is more selective. It reduces false alarms to 2,190, but detects only 442 failures and misses 735. Gradient Boosting lies between these two patterns, with 695 detected failures

and 4,563 false alarms. In practical terms, Logistic Regression gives broad screening coverage, Random Forest gives fewer alerts, and Gradient Boosting provides a middle position. Which pattern is preferable depends on the relative cost of a missed failure and a false alarm. Those costs are not estimated in this project.

The validation threshold analysis leads to the same general lesson. For each model, thresholds chosen to reach at least 80% recall label a much larger share of observations as failed than thresholds selected to maximize F1. Thus, recall can be increased, but only by accepting lower precision and more firms sent for further review. Since these cut-offs were studied on validation data, they are not used to revise the final-test results above.

5.3 Precision–recall analysis

Figure 4 shows the precision–recall curves for the three key models. The dashed horizontal line represents the failed-class share in the test data. A model needs to remain above this reference to provide more useful ranking than a non-informative classifier at the corresponding recall level.

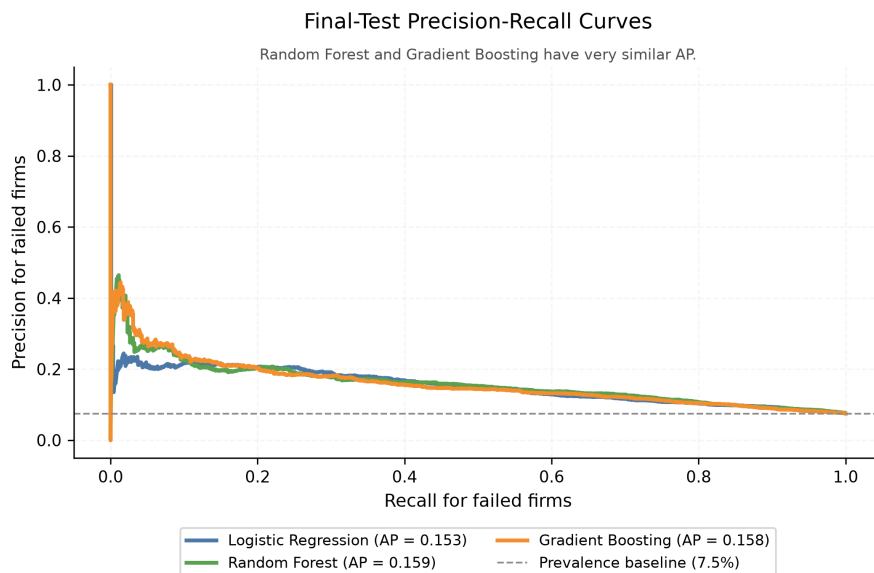


Figure 4: Final-test precision–recall curves for the key models. The legend reports average precision, labelled PR-AUC in the project outputs.

The curves are close, which is consistent with their similar average precision values of 0.153, 0.159, and 0.158. Random Forest has the numerically highest value, but its advantage over the validation-selected Gradient Boosting model is very small. Logistic Regression is also close in ranking performance even though its classifications at the default threshold produce a very different confusion matrix. This illustrates why a model’s ranking quality and its decisions at one threshold should be discussed separately. Overall, the models contain predictive information, but the low precision levels also show that bankruptcy prediction remains difficult.

6 Model Interpretation

6.1 Logistic Regression coefficients

Figure 5 shows the largest coefficients from the main Logistic Regression. Because the predictors were standardized before fitting, their coefficient magnitudes can be compared on a common scale. A positive coefficient is associated with higher predicted log-odds of failure, while a negative coefficient is associated with lower predicted log-odds, holding the other predictors fixed.

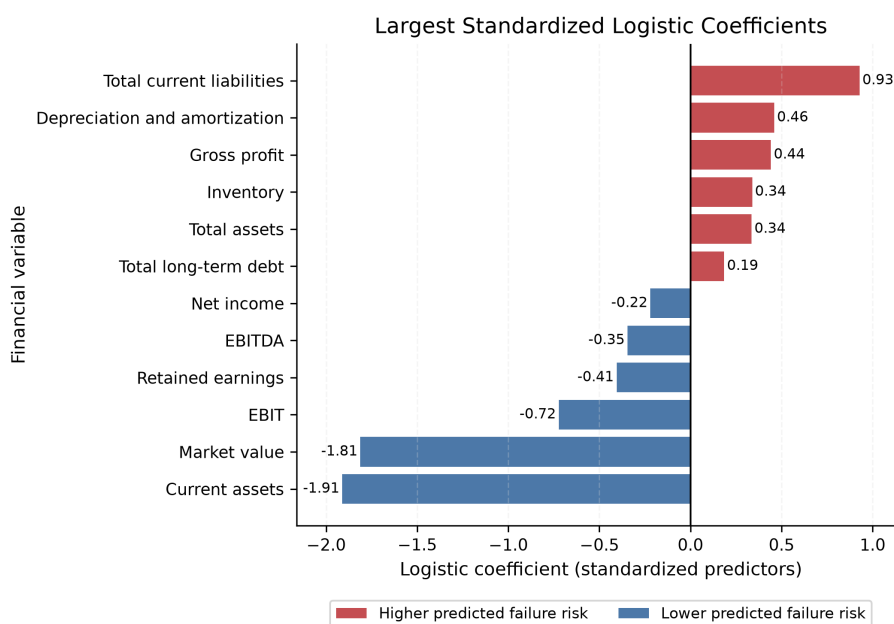


Figure 5: Largest Logistic Regression coefficients. Predictors are standardized, and the target value one represents a failed observation.

Current assets and market value have the largest negative coefficients, at -1.912 and -1.812 . EBIT, retained earnings, and EBITDA also have negative coefficients among the variables shown. In this fitted model, higher values of these variables are associated with lower predicted failure risk. This is broadly consistent with the descriptive results: failed observations have lower median current assets, market value, net income, and retained earnings.

Total current liabilities have the largest positive coefficient at 0.928 . Depreciation and amortization, gross profit, inventory, and total assets also have positive coefficients among the largest values. These signs must be read carefully. The model includes several related accounting totals at the same time, so a coefficient describes a conditional relationship after the other variables are held fixed. For example, a positive gross-profit coefficient does not mean that higher gross profit causes bankruptcy. It may reflect the combination of correlated size, revenue, cost, and balance-sheet variables in the fitted prediction model.

The coefficients help explain how the model forms its scores, but they are not causal estimates. The analysis does not use a causal research design, and it does not report statistical inference for individual coefficients. Their role here is predictive interpretation.

6.2 Tree-based feature importance

Figure 6 reports the leading impurity-based feature importance values for Random Forest and Gradient Boosting. A larger value means that a variable contributed more to reducing impurity across the fitted tree splits within that model. Unlike a logistic coefficient, this measure does not show whether higher values increase or decrease predicted risk.

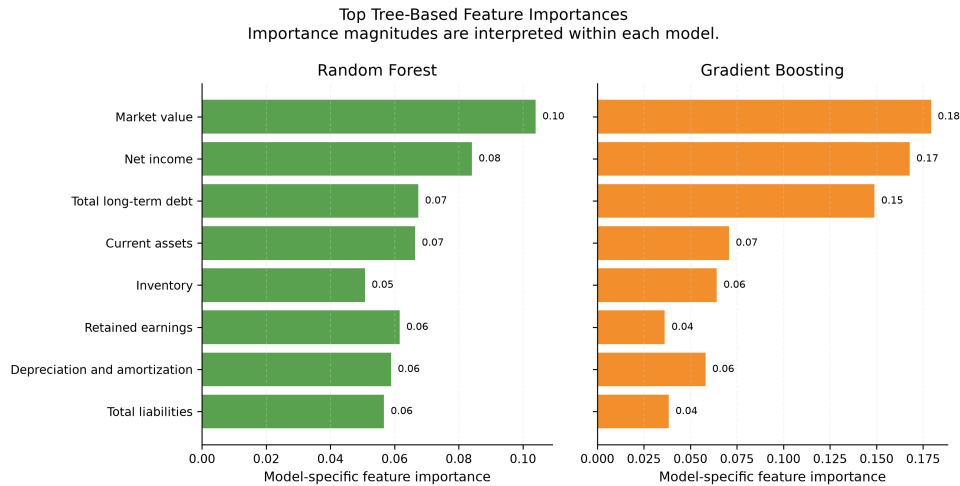


Figure 6: Leading feature importance values within the Random Forest and Gradient Boosting models. Magnitudes should be compared mainly within each model rather than directly across models.

Market value, net income, and total long-term debt are the three most important features in both ensembles. Current assets also appear near the top, followed by variables such as retained earnings, inventory, depreciation and amortization, and total liabilities. This overlap suggests that market valuation, profitability, liquidity, and debt information all contribute to the models' predictions. It does not establish that these variables cause failure.

The importance profiles also differ in concentration. Random Forest spreads importance more evenly across the predictors, while Gradient Boosting assigns a larger share to its first few variables. The absolute values should not be compared as if they were on one common scale, because each ensemble constructs and combines trees differently. Impurity-based importance can also favour variables that provide many useful split points and can divide importance among correlated predictors. The figure is therefore best used as a summary of the fitted models, not as a definitive ranking of economic causes.

Taken together, the two interpretation methods offer different information. Logistic Regression provides signed and relatively direct relationships on the log-odds scale. The ensembles allow nonlinear patterns and interactions, but their importance values are less direct. This is the central interpretability trade-off in the project: more flexible models can represent richer patterns, while the logistic model is easier to explain in financial terms.

7 PCA Results

The PCA extension asks whether the 18 standardized financial variables can be compressed without losing much predictive information. The first few components explain a large share of the variation in the predictors. Two components retain 83.8% of total variance, five retain 93.9%,

and ten retain 99.2%. The full explained-variance curve is provided in Appendix C.

High explained variance does not automatically produce good failure prediction. Table 3 reports the validation results for PCA followed by Logistic Regression. With only two components, ROC-AUC is 0.498 and PR-AUC is 0.063, even though those components retain most of the predictor variance. Performance improves as more components are included. The ten-component version has the highest failed-class F1-score, at 0.152, while the twelve-component version has the highest PR-AUC, at 0.151. These differences are small.

Table 3: Internal validation results for PCA with Logistic Regression

Components	Explained variance	ROC-AUC	PR-AUC	Failed F1
2	0.838	0.498	0.063	0.130
3	0.892	0.642	0.116	0.146
5	0.939	0.651	0.124	0.148
8	0.980	0.669	0.140	0.150
10	0.992	0.671	0.147	0.152
12	0.998	0.674	0.151	0.150
15	1.000	0.671	0.147	0.149
18	1.000	0.671	0.147	0.149

For comparison, Logistic Regression on the original standardized predictors has a validation PR-AUC of 0.148 and a failed-class F1-score of 0.149. The PCA versions with ten or twelve components are therefore very similar, rather than clearly better. PCA can reduce the input dimension, but substantial compression to two or three components removes information that matters for separating failed and alive observations.

PCA also weakens financial interpretation. Each component is a weighted combination of many accounting variables, so it cannot be discussed as directly as current assets, market value, income, or debt. In this project, the small change in validation performance does not provide a strong reason to accept that loss of interpretability. The PCA exercise is still useful because it demonstrates an important distinction: variation in the predictors is not the same as variation that helps predict the target. All PCA comparisons are validation results; they are not additional final-test claims.

8 Critical Reflection and Limitations

The project has several useful design choices. Splitting by company prevents the same firm from appearing in both training and test data. Keeping the test set separate from model and threshold selection also makes the final comparison more credible. Finally, the majority baseline and failed-class metrics expose weaknesses that would be hidden by accuracy. These choices improve the analysis, but they do not remove the following limitations.

8.1 Data and target limitations

The data contain repeated company-year observations, and observations from one firm are likely to be related over time. The company-level split handles direct firm overlap, but the design is not a true forecasting exercise because the training and test samples cover overlapping calendar

years. It therefore tests generalization to unseen companies within a similar historical period, not performance in a later economic period. The changing annual failure rate also suggests that time and sample composition matter.

The failed/alive label is useful for binary classification, but the operational prediction horizon is not fully defined. A real early-warning application would need a statement such as “failure within the next twelve months” and would need to confirm that every accounting variable was available before that forecast date. Without this timing detail, the results should be read as classification of company-year observations rather than a complete real-time forecasting system.

The predictors are financial statement levels. They contain information about firm condition, but they may also partly reflect company size and accounting scale. This matters because large firms naturally have larger assets, liabilities, sales, and expenses. The project does not add ratios, changes over time, industry indicators, macroeconomic conditions, governance information, or market returns. Dataset coverage and reporting choices may also make the sample different from other countries, private firms, or later periods. These issues limit external generalization.

8.2 Validation and modelling limitations

The reported results come from one grouped train–test split and one internal validation split. They do not show how much performance would vary under other valid splits. There are no confidence intervals, repeated grouped validation, or statistical tests of the small differences between models. In particular, the very close PR-AUC values should not be treated as proof that one ensemble is generally superior.

Hyperparameter searches are deliberately limited, which keeps the course project understandable but may leave better settings unexplored. In addition, the saved final models are fitted on the inner model-training portion and are not refitted after validation. This gives a conservative final test because some available training observations are not used in the final fit, but it is not the most data-efficient workflow. A later version could lock all choices, refit on the complete non-test sample, and then evaluate once on a still-unused test set.

The models evaluate discrimination and classification, not probability calibration. Their outputs should therefore not automatically be interpreted as literal bankruptcy probabilities. Logistic coefficients can also change when correlated predictors are included together, while impurity-based tree importance has its own biases. Both are descriptions of fitted predictive models, not causal evidence.

8.3 Decision and practical limitations

The threshold analysis makes the precision–recall trade-off visible, but the project does not estimate the monetary cost of a missed failure or a false alarm. It also does not specify how many flagged firms an analyst could review. Without a loss function or capacity constraint, there is no single economically optimal model or threshold. The low failed-class precision is especially important: even the more selective models produce many false alerts.

A real financial application would require further checks, including probability calibration, chronological backtesting, stability across industries and economic conditions, stress testing, fairness and governance review, and a plan for monitoring model drift. Human review would

also be necessary before any credit, investment, or regulatory action. The current project is a reproducible educational study and screening exercise; it is not production-ready.

8.4 Priorities for future work

The first improvement would be to define a clear forward failure horizon and use rolling or chronological evaluation. Repeated company-grouped validation and uncertainty intervals would show whether model differences are stable. Financial ratios and trends could reduce the influence of firm size, while industry and macroeconomic variables could add context. After model selection, probability calibration should be tested, and the threshold should be chosen using explicit financial costs and review capacity. These steps would improve decision relevance more than simply adding another complex algorithm.

9 Conclusion

This paper examined how well company-year financial statement variables can identify failed observations for companies not seen during model fitting. It also tested whether flexible tree-based classifiers improve on an interpretable Logistic Regression benchmark. The results show that the financial variables contain information for out-of-company generalization, but bankruptcy classification remains difficult and no model is best for every objective.

The majority-class baseline achieved 92.5% accuracy while detecting none of the failed observations. This is the clearest reason not to judge the project by accuracy alone. Among the key models, Random Forest produced the numerically highest observed final-test PR-AUC and failed-class F1-score, while Gradient Boosting was the validation-selected ranking model. This distinction avoids choosing a model after seeing the test set. Gradient Boosting achieved the highest balanced accuracy and detected more failures than Random Forest, but it also generated more false alarms. Logistic Regression detected the largest share of failures and was the easiest model to explain, although its broad screening rule produced many false positives. The tree-based models therefore improved some measures, but they did not dominate the logistic benchmark on every relevant outcome.

The additional analyses support this trade-off-based view. Threshold tuning showed that higher recall requires lower precision and a larger review burden. The PCA extension retained much of the variation in fewer dimensions, but it did not provide a clear predictive advantage and made the financial meaning of the inputs less direct. Logistic coefficients and tree importance highlighted market value, profitability, liquidity, and debt variables, but these patterns are predictive associations rather than causal effects.

The main lesson is that company-year bankruptcy-risk classification is both a statistical and financial decision problem. Model rankings, classification thresholds, error costs, and interpretability must be considered together. Under the current company-level split, the models provide a structured way to rank or screen firms, not a basis for automatic decisions. A stronger next study would define a forward prediction horizon, use chronological testing, assess calibration and uncertainty, and select thresholds using explicit financial costs. Within its educational scope, the project demonstrates a reproducible and careful application of classification methods to an important problem in finance.

References

- Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4):589–609.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Davis, J. and Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 233–240. Association for Computing Machinery.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232.
- James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. (2023). *An Introduction to Statistical Learning: With Applications in Python*. Springer, Cham.
- Jolliffe, I. T. and Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065):20150202.
- Lombardo, G., Pellegrino, M., Adosoglou, G., Cagnoni, S., Pardalos, P. M., and Poggi, A. (2022). Machine learning for bankruptcy prediction in the american stock market: Dataset and benchmarks. *Future Internet*, 14(8):244.
- Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research*, 18(1):109–131.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., VanderPlas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, É. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12(85):2825–2830.

A Additional Model Results

Table 4: Model specifications used in the final-test comparison

Model	Specification
Majority baseline	Always predicts the most common class
Logistic Regression	L2 penalty, $C = 1.0$, balanced class weights
L1 Logistic Regression	L1 penalty, $C = 0.1$, balanced class weights
L2 Logistic Regression	L2 penalty, $C = 0.1$, balanced class weights
Decision Tree	Maximum depth 5; minimum leaf size 50
Random Forest	300 trees; unrestricted depth; minimum leaf size 50; $\sqrt{p}$ candidate predictors per split
Gradient Boosting	150 trees; learning rate 0.05; maximum depth 3; minimum leaf size 100

Table 5 gives the complete final-test comparison, including the regularized logistic models and the single Decision Tree. The regularized logistic variants are very close to the fixed Logistic Regression benchmark. The Decision Tree improves failed recall over Random Forest but has weaker ranking performance and produces more false positives. The source is `outputs/tables/final_test_metrics.csv`.

Table 5: Complete final-test model comparison

Model	Accuracy	Balanced acc.	ROC-AUC	PR-AUC	Failed precision	Failed recall	Failed F1
Majority baseline	0.925	0.500	0.500	0.075	0.000	0.000	0.000
Logistic Regression	0.360	0.601	0.690	0.153	0.095	0.884	0.172
L1 Logistic Regression	0.358	0.602	0.692	0.154	0.095	0.889	0.172
L2 Logistic Regression	0.358	0.602	0.692	0.153	0.095	0.888	0.172
Decision Tree	0.600	0.581	0.633	0.122	0.102	0.558	0.173
Random Forest	0.814	0.612	0.699	0.159	0.168	0.376	0.232
Gradient Boosting	0.679	0.638	0.688	0.158	0.132	0.590	0.216

Figure 7 provides a supplementary ROC-curve view for the three key predictive models. ROC curves are useful for comparing threshold-independent ranking performance, and the diagonal line represents a random classifier. However, because failed firms are rare in this application, the main discussion emphasizes precision–recall performance and minority-class operating metrics. Those measures show the screening burden more directly than the false-positive rate alone.

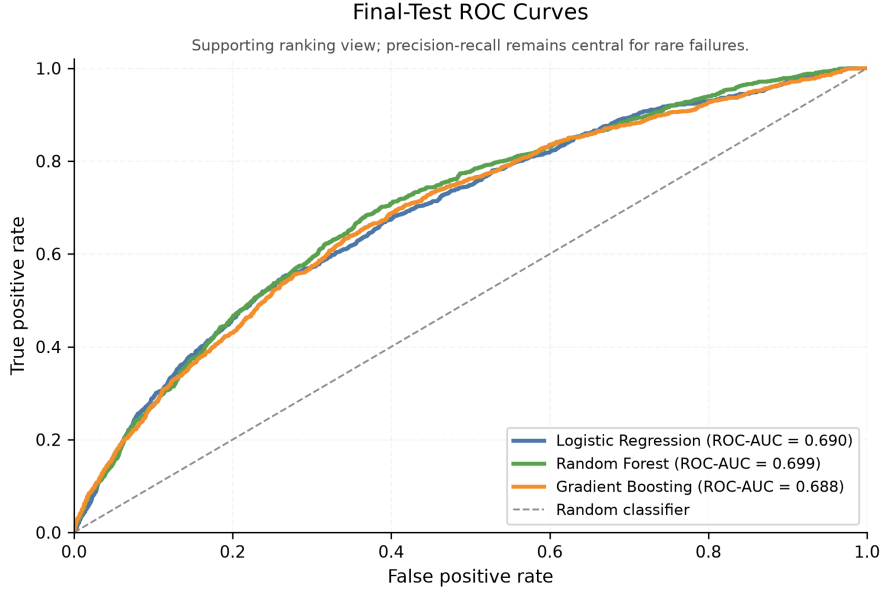


Figure 7: Supplementary final-test ROC curves for the key models. The legend reports ROC-AUC values. The figure is included as a secondary ranking view; the main analysis emphasizes precision–recall performance because failed firms are rare.

B Threshold Analysis

Table 6 reports the validation thresholds discussed in Section 4. For each rule, the table also reports the share of validation observations that would be labelled failed. The sharp rise in this share under the 80% recall rule shows the review burden created by a high failure-detection target. These settings are descriptive validation choices and were not applied to the final test comparison.

Table 6: Selected thresholds from the internal validation sample

Model	Rule	Threshold	Precision	Recall	F1	Predicted failed share
Gradient Boosting	Maximum F1	0.68	0.212	0.220	0.216	0.068
	Recall ≥ 0.60	0.48	0.109	0.613	0.185	0.370
	Recall ≥ 0.80	0.37	0.090	0.823	0.162	0.599
Logistic Regression	Maximum F1	0.53	0.190	0.320	0.239	0.110
	Recall ≥ 0.60	0.52	0.088	0.682	0.156	0.507
	Recall ≥ 0.80	0.50	0.082	0.846	0.149	0.678
Random Forest	Maximum F1	0.57	0.187	0.265	0.219	0.093
	Recall ≥ 0.60	0.37	0.115	0.616	0.194	0.351
	Recall ≥ 0.80	0.27	0.100	0.812	0.178	0.532

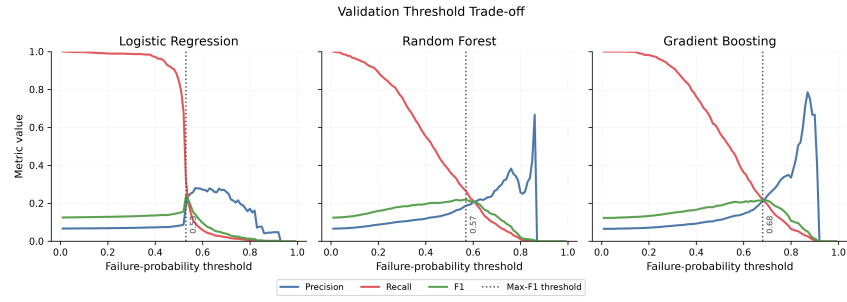


Figure 8: Validation precision, recall, and failed-class F1 across candidate thresholds. The figure uses validation predictions only.

C PCA Diagnostics

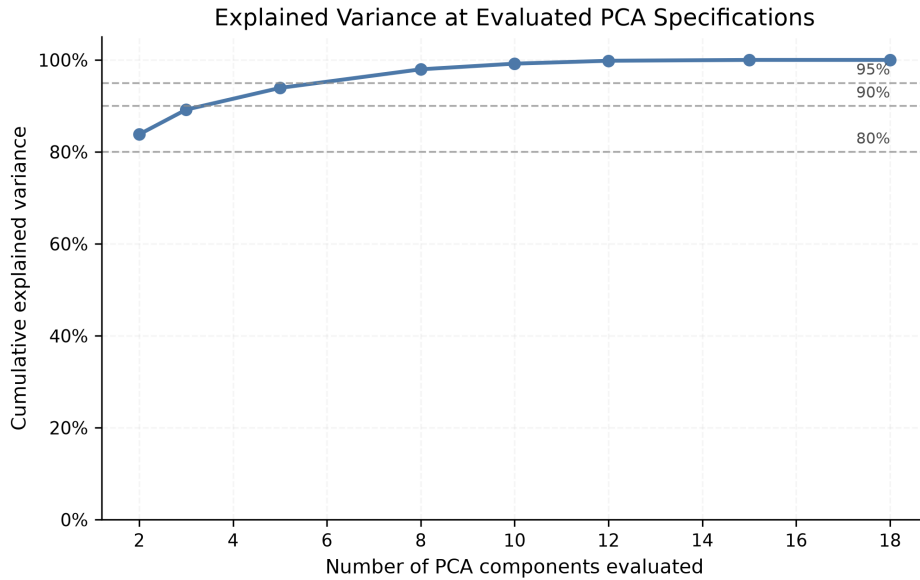


Figure 9: Cumulative explained variance from PCA on the standardized financial predictors. Results use the internal validation design.

Figure 10 complements the PCA table in Section 7. It shows that classification performance rises as the first components are added and then changes little. The full numerical results are stored in `outputs/tables/pca_logistic_results.csv`, while component weights are stored in `outputs/tables/pca_component_loadings.csv`.

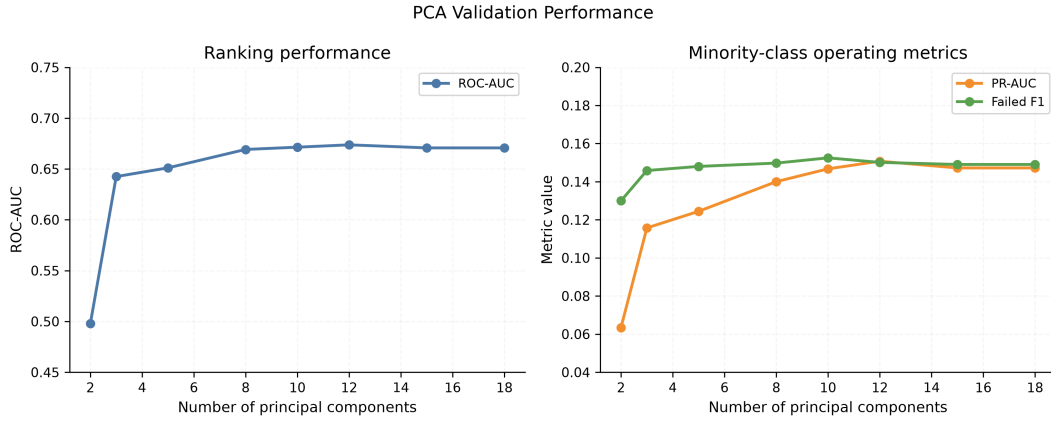


Figure 10: Internal validation metrics for PCA plus Logistic Regression across the tested component counts.

D Reproducibility and Code Structure

The project uses a task-based workflow and fixed random seeds. Configuration, paths, feature names, and the split seed are defined in `src/bankruptcy_ml/config.py`. The remaining modules separate data validation, exploratory analysis, preprocessing, company-level splitting, model fitting, evaluation, threshold analysis, PCA, interpretation, and visualization. Files in `tasks/` connect these stages into the build workflow, and files in `tests/` check the main data and modelling behaviour.

From the repository root, the complete existing workflow is run with:

```
pixi run build
pixi run test
```

The build command creates the processed data, fitted model files, result tables, and figures through the declared task dependencies. The test command runs the automated test suite. Generated analysis artifacts are kept under `data/processed/` and `outputs/`. The paper-ready layer under `outputs/paper/` contains simplified copies and summaries derived from the existing results; it does not represent a second model run.

The main evidence files used in this paper are listed below. Keeping these tables alongside the figures makes every reported value traceable to a saved project output.

- `final_test_metrics.csv` and `final_test_classification_reports.csv`;
- `selected_thresholds.csv`;
- `logistic_coefficients.csv` and `tree_feature_importance.csv`;
- `pca_logistic_results.csv`.